

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSECODE: CSA1603
COURSE NAME: Data Warehousing and Data Mining
NAME: P. GAYATHRI
REG.NO: 192425277

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

*I P. GAYATHRI/192425277 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: P. GAYATHRI

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

Q1. Dataset: Hospital Patient Records

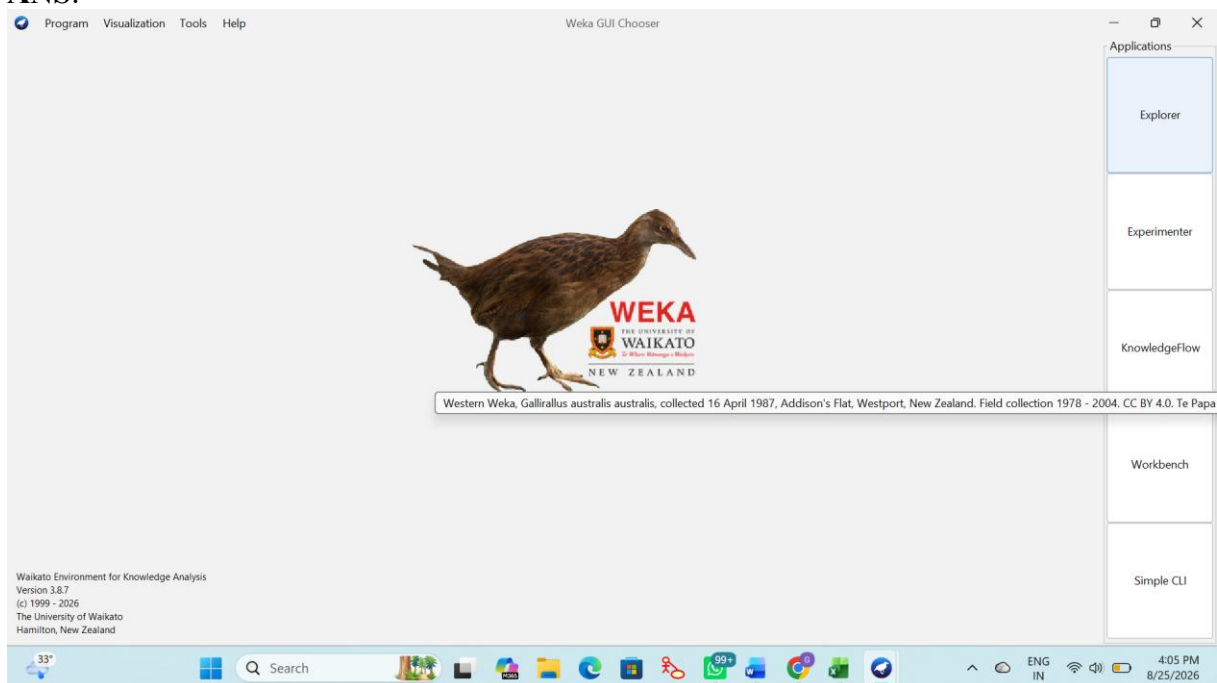
PID Symptoms Observed

- P1 Fever, Cough
- P2 Fever, Headache
- P3 Cough, Headache
- P4 Fever, Cough, Headache
- P5 Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints. (BL4 – 1.2 Marks)

ANS:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: hospital-weka.filters.unsupervised.attribute.Remove-R1
Instances: 5 Attributes: 4 Sum of weights: 5

Attributes: All None Invert Pattern

No.	Name
1	☒ Fever
2	☐ Cough
3	☐ Headache
4	☐ Fatigue

Remove

Status: OK

Selected attribute: Name: Fever Missing: 0 (0%) Distinct: 2 Type: Numeric Unique: 1 (20%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.8
StdDev	0.447

Class: Fatigue (Num) Visualize All

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **NumericToNominal -R first-last** Apply Stop

Current relation: hospital-weka.filters.unsupervised.attribute.Remove-R1-weka.filters.unsupervised...
Instances: 5 Attributes: 4 Sum of weights: 5

Attributes: All None Invert Pattern

No.	Name
1	☒ Fever
2	☐ Cough
3	☐ Headache
4	☐ Fatigue

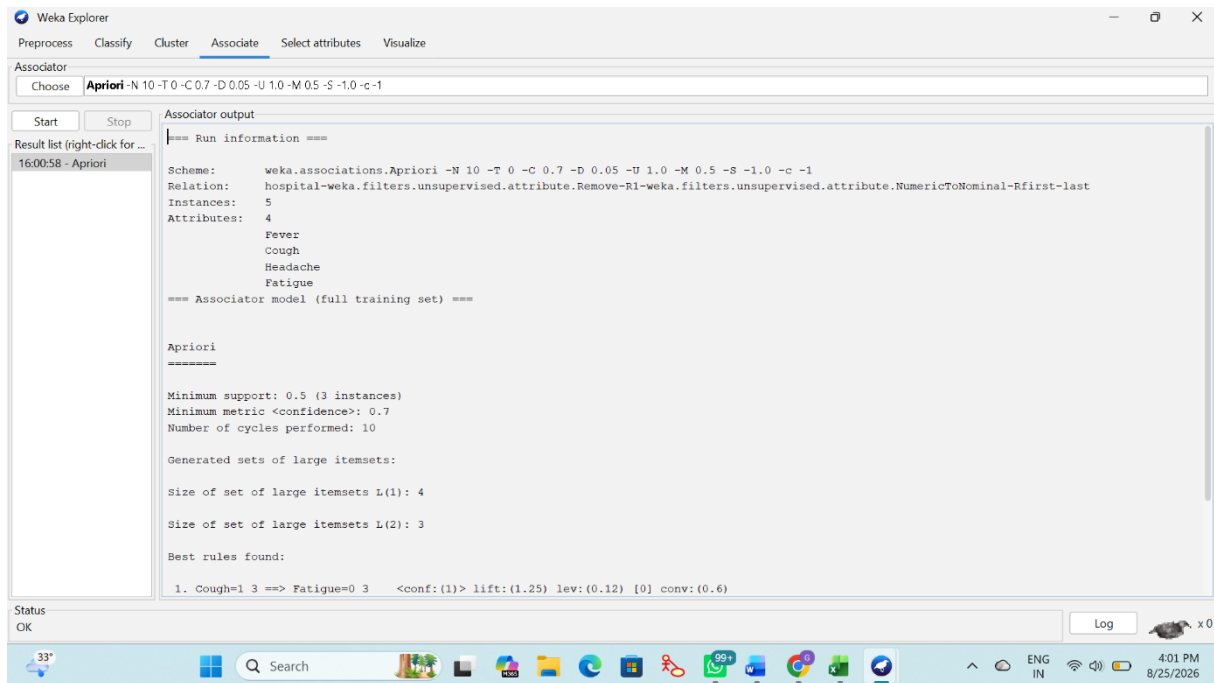
Remove

Status: OK

Selected attribute: Name: Fever Missing: 0 (0%) Distinct: 2 Type: Nominal Unique: 1 (20%)

No.	Label	Count	Weight
1	0	1	1
2	1	4	4

Class: Fatigue (Nom) Visualize All



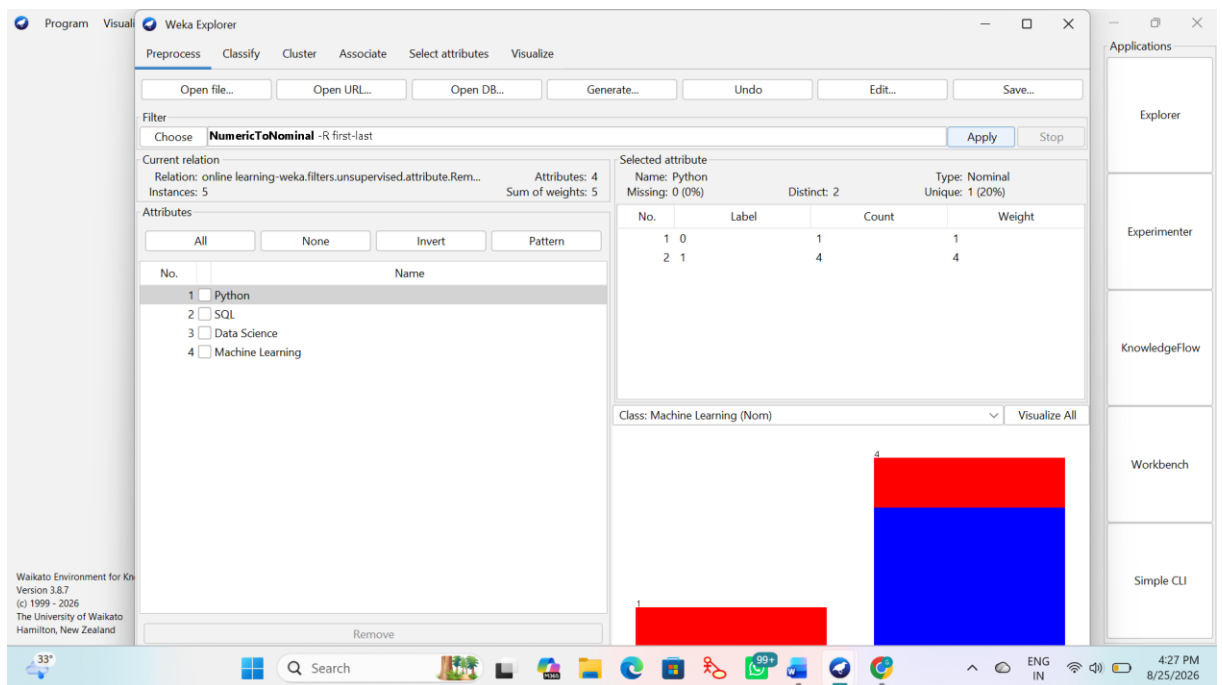
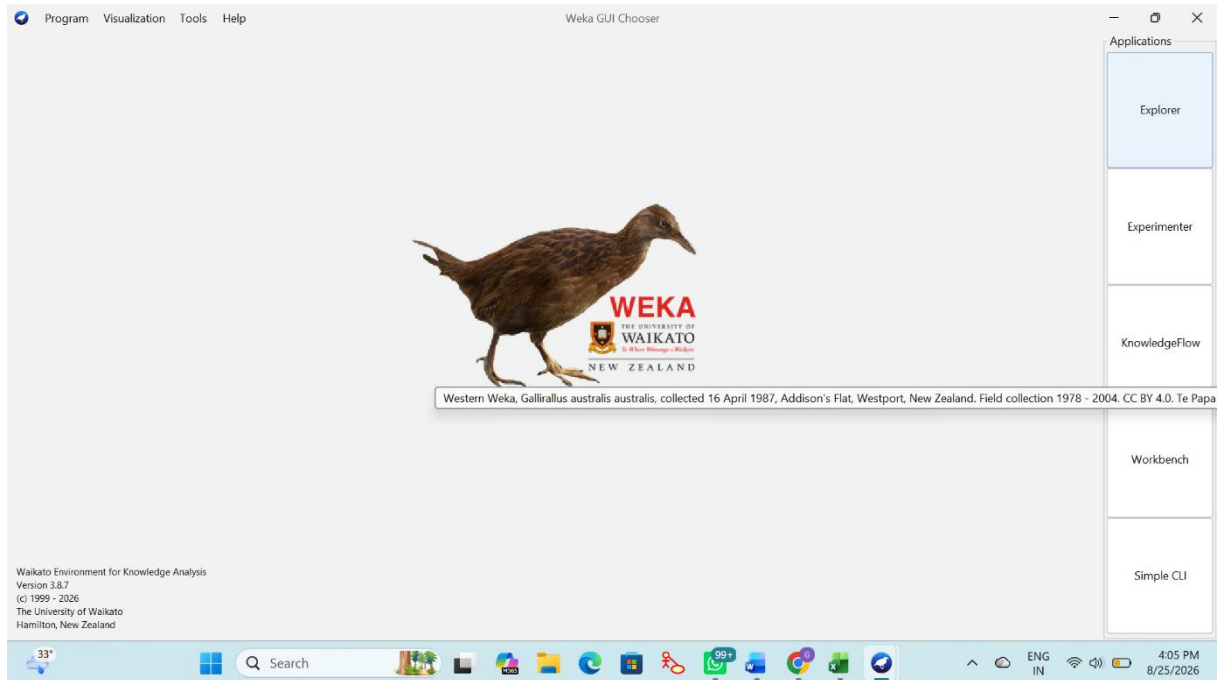
No association rules containing Fever satisfy the minimum support of 50% and minimum confidence of 70%. Therefore, there are no valid rules under the given constraint.

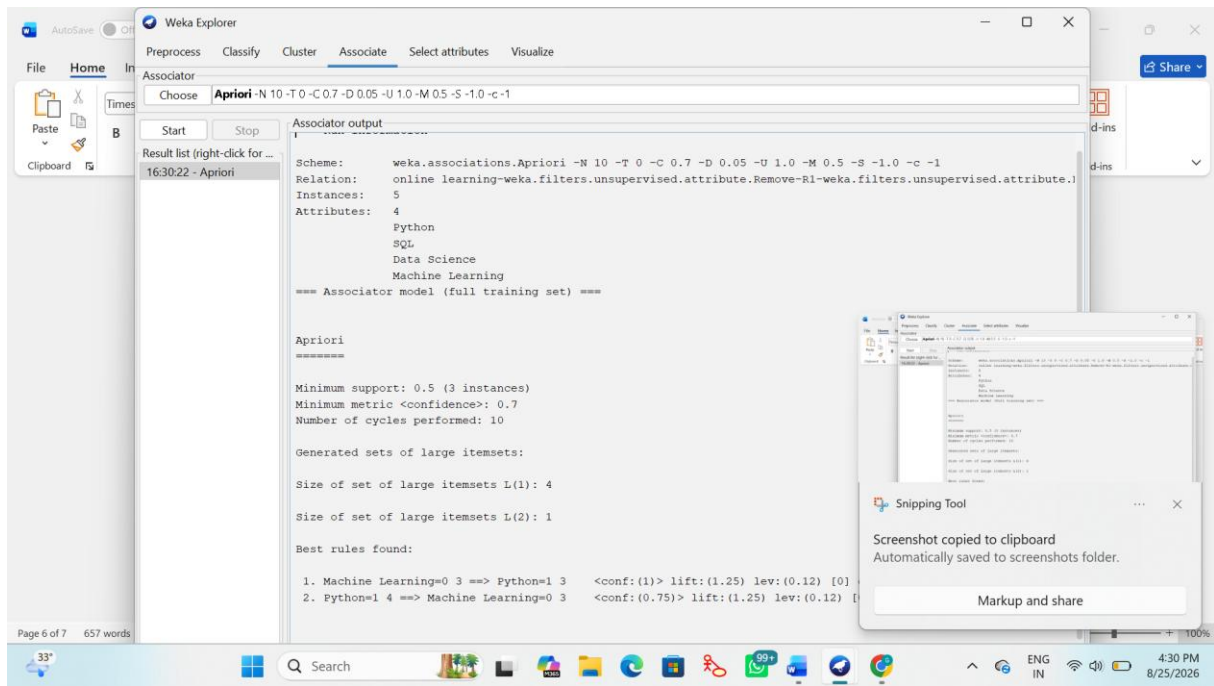
Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining. (BL4 – 1.2 Marks)



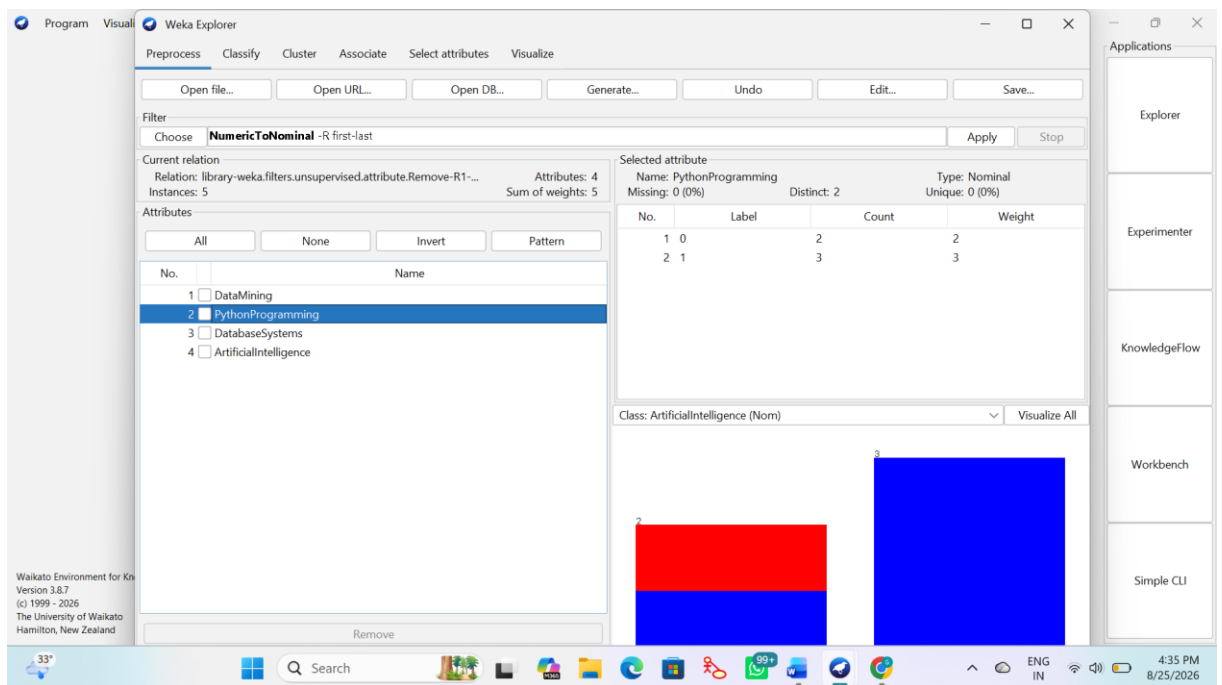
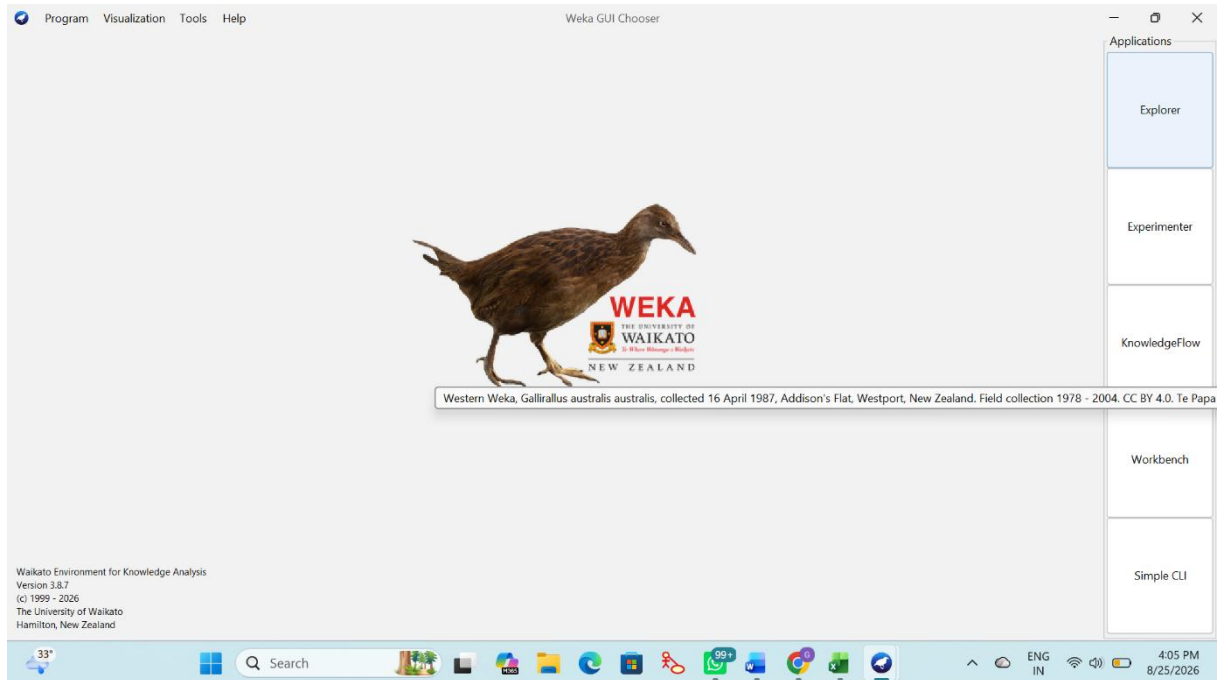


Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation. (BL5 – 1.2 Marks)



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split % 66

☐ Classes to clusters evaluation (Nom) ArtificialIntelligence

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

16:36:23 - EM

Cluster output

```
DataMining
0 2
1 5
[total] 7
PythonProgramming
0 3
1 4
[total] 7
DatabaseSystems
0 3
1 4
[total] 7
ArtificialIntelligence
0 5
1 2
[total] 7

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0 5 (100%)

Log likelihood: -2.38884
```

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Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associate

Choose **Apriori** -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

16:36:45 - Apriori

16:37:12 - Apriori

Associator output

```
Attributes: 4
DataMining
PythonProgramming
DatabaseSystems
ArtificialIntelligence

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.5 (3 instances)
Minimum metric <confidence>: 0.7
Number of cycles performed: 10

Generated sets of large itemsets:

Size of set of large itemsets L(1): 4

Size of set of large itemsets L(2): 3

Best rules found:

1. PythonProgramming=1 3 ==> ArtificialIntelligence=0 3 <conf: (1)> lift: (1.25) lev: (0.12) [0] conv: (0.6)
2. DatabaseSystems=1 3 ==> ArtificialIntelligence=0 3 <conf: (1)> lift: (1.25) lev: (0.12) [0] conv: (0.6)
3. ArtificialIntelligence=0 4 ==> DataMining=1 3 <conf: (0.75)> lift: (0.94) lev: (-0.04) [0] conv: (0.4)
4. DataMining=1 4 ==> ArtificialIntelligence=0 3 <conf: (0.75)> lift: (0.94) lev: (-0.04) [0] conv: (0.4)
5. ArtificialIntelligence=0 4 ==> PythonProgramming=1 3 <conf: (0.75)> lift: (1.25) lev: (0.12) [0] conv: (0.8)
6. ArtificialIntelligence=0 4 ==> DatabaseSystems=1 3 <conf: (0.75)> lift: (1.25) lev: (0.12) [0] conv: (0.8)
```

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Q4. Dataset: University Course Hierarchy

Student ID

Courses Taken

S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "**Computer Science**" level. Explain how the course hierarchy affects the discovery of association rules. (BL5 – 1.2 Marks)

The screenshot shows the Weka GUI Explorer interface. The 'Preprocess' tab is active, displaying a list of attributes: Engineering, ComputerScience, MachineLearning, and InformationTechnology. The 'Selected attribute' table shows the distribution of these attributes. A bar chart visualizes the data for the 'InformationTechnology' class, showing a red bar for 'Engineering' and a blue bar for 'ComputerScience'.

No.	Label	Count	Weight
1	1	5	5

Class: InformationTechnology (Nom) Visualize All

Bar chart showing the distribution of the 'InformationTechnology' class. The x-axis represents the class, and the y-axis represents the count. The bar is divided into two segments: a red segment for 'Engineering' and a blue segment for 'ComputerScience'.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **ZeroR**

Test options

☒ Use training set

☐ Supplied test set Set...

☐ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) InformationTechnology

Start Stop

Result list (right-click for options)

16:45:01 - rules.ZeroR

Classifier output

Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	3	60	%
Incorrectly Classified Instances	2	40	%
Kappa statistic	0		
Mean absolute error	0.4857		
Root mean squared error	0.4907		
Relative absolute error	100	%	
Root relative squared error	100	%	
Total Number of Instances	5		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
1.000	1.000	0.600	1.000	0.750	?	?	0.500	0.600	0
0.000	0.000	?	0.000	?	?	?	0.500	0.400	1
Weighted Avg.	0.600	0.600	?	0.600	?	?	0.500	0.520	

=== Confusion Matrix ===

a b <-- classified as

3 0 | a = 0

2 0 | b = 1

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

16:45:33 - Apriori

Associator output

Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1

Relation: university courses-weka.filters.unsupervised.attribute.Remove-R1-weka.filters.unsupervised.attribute

Instances: 5

Attributes: 4

Engineering

ComputerScience

MachineLearning

InformationTechnology

=== Associator model (full training set) ===

Apriori

=====

Minimum support: 0.7 (3 instances)

Minimum metric <confidence>: 0.7

Number of cycles performed: 6

Generated sets of large itemsets:

Size of set of large itemsets L(1): 4

Size of set of large itemsets L(2): 4

Size of set of large itemsets L(3): 1

Best rules found:

1. ComputerScience=1 3 ==> Engineering=1 3 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
2. MachineLearning=0 3 ==> Engineering=1 3 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
3. InformationTechnology=0 3 ==> Engineering=1 3 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
4. InformationTechnology=0 3 ==> ComputerScience=1 3 <conf:(1)> lift:(1.67) lev:(0.24) [1] conv:(1.2)

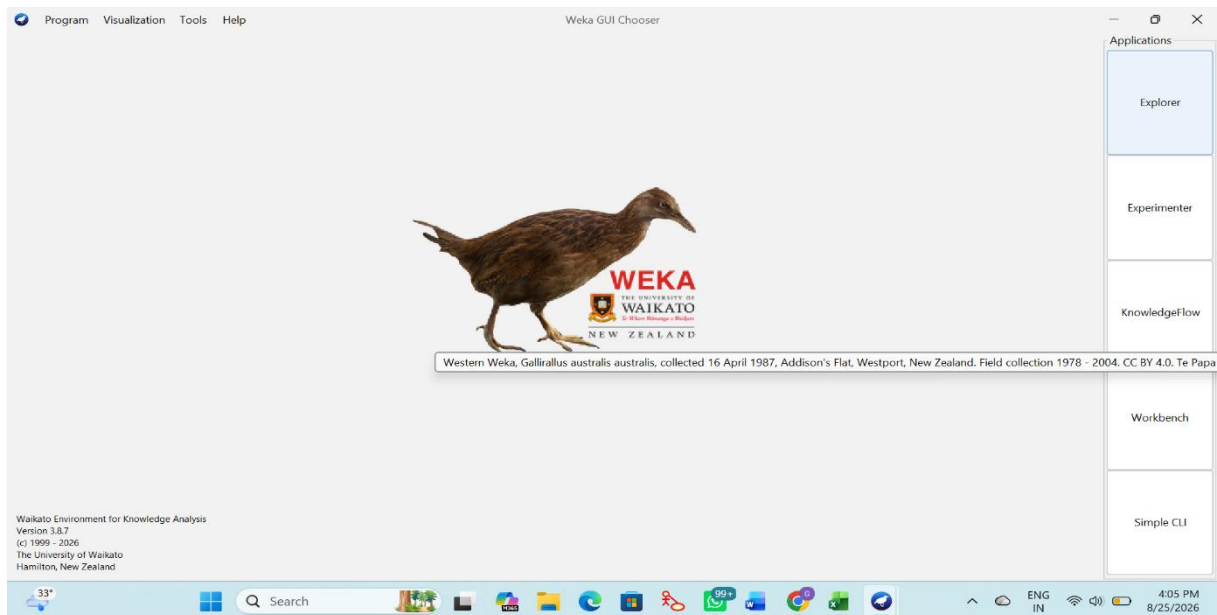
Q5. Dataset: Insurance Customer Records

CID Age Group Income Level Policy Purchased

C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. (*BL4 – 1.2 Marks*)



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **NumericToNominal -R first-last** Apply Stop

Current relation: insurance-weka.filters.unsupervised.attribute.Remove-R... Attributes: 3 Sum of weights: 5
Instances: 5

Attributes: All None Invert Pattern

No.	Name
1	AgeGroup
2	IncomeLevel
3	PolicyPurchased

Selected attribute: Name: AgeGroup Missing: 0 (0%) Distinct: 3 Type: Nominal Unique: 2 (40%)

No.	Label	Count	Weight
1	Young	3	3
2	Middle	1	1
3	Senior	1	1

Class: PolicyPurchased (Nom) Visualize All

Waikato Environment for Knowledge Mining
Version 3.8.7
(c) 1999 - 2026
The University of Waikato
Hamilton, New Zealand

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Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

File Home In

AutoSave On

Clipboard

Associate

Choose **Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1**

Start Stop

Result list (right-click for ...)

16:53:09 - Apriori

Associator output

```
=== Run information ===  
  
Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1  
Relation: insurance-weka.filters.unsupervised.attribute.Remove-R1-weka.filters.unsupervised.attribute.NumericToNominal  
Instances: 5  
Attributes: 3  
AgeGroup  
IncomeLevel  
PolicyPurchased  
  
=== Associator model (full training set) ===  
  
No large itemsets and rules found!
```

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